Live Gold & Silver Price Prediction using Machine Learning

A Mini Project in Machine Learning

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Introduction

- Gold and silver are the two most tracked precious metals, priced in US dollars per troy ounce.
- Their prices move with demand, inflation, the dollar and global market sentiment.
- This project fetches the LIVE price of both metals and converts it into Indian Rupees per gram.
- A Machine Learning model then forecasts the price for the next 12 months.
- Everything is shown on one web dashboard, so the numbers can be read without any training.

Problem Statement

- Given the historical monthly price of gold and silver, predict their price for the coming months.
- This is a supervised regression problem: past behaviour is the input, future price is the output.
- Several models are trained, scored on data they have never seen, and the best one is kept.
- The winning model is connected to a live price feed and presented on a dashboard in rupees.

Objectives

- Build a clean monthly price dataset for gold and silver from the year 2000 up to today.
- Read the live price of both metals and the US dollar to rupee rate from public APIs.
- Engineer features that capture short-term momentum instead of raw price levels.
- Compare Random Forest, Linear Regression and Ridge against a naive baseline.
- Deliver an attractive dashboard showing present and future prices in Indian Rupees.

Live Data Sources

- api.gold-api.com/price/XAU - live gold price in US dollars per troy ounce.
- api.gold-api.com/price/XAG - live silver price in US dollars per troy ounce.
- open.er-api.com - today's US dollar to Indian rupee exchange rate.
- All three are free public feeds, so no key or login is needed.
- Conversion: Rs per gram = US$ per ounce x USD-INR rate / 31.1035 (one troy ounce = 31.1035 g).

Dataset

- Two files: gold_prices.csv and silver_prices.csv - one row per month since January 2000.
- Columns: Date, Year, Month and Price (US dollars per troy ounce).
- Yearly anchors are approximate LBMA annual averages, interpolated month to month with momentum.
- Model features: the last 3 monthly returns, plus the month and the year.
- The live API price is appended as the latest point, so the forecast always starts from today.

Methodology

- The model predicts the month-to-month RETURN, never the raw price.
- Why: a tree model cannot guess a price higher than anything it saw, so a 25 year rising trend breaks it.
- Price is rebuilt as next price = last price x (1 + predicted return).
- Forecasting rolls forward one month at a time, feeding each answer back as the next input.
- The split is chronological - the last 60 months are held out and never used for fitting.

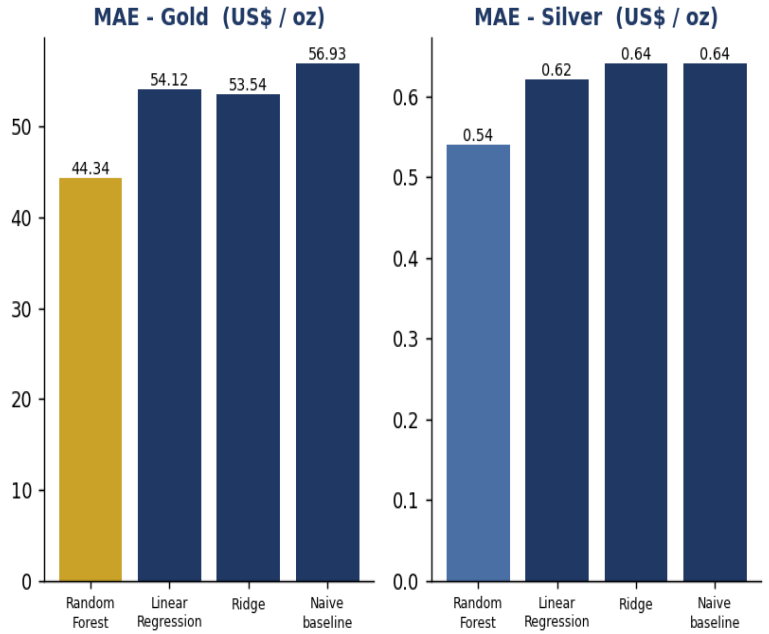
Technology Used

- Language: Python.
- Data handling: pandas and NumPy.
- Models: scikit-learn - Random Forest, Linear Regression, Ridge.
- Backend: Flask, serving one JSON summary to the page.
- Frontend: Chart.js on a dark dashboard. Model save / load: Joblib.

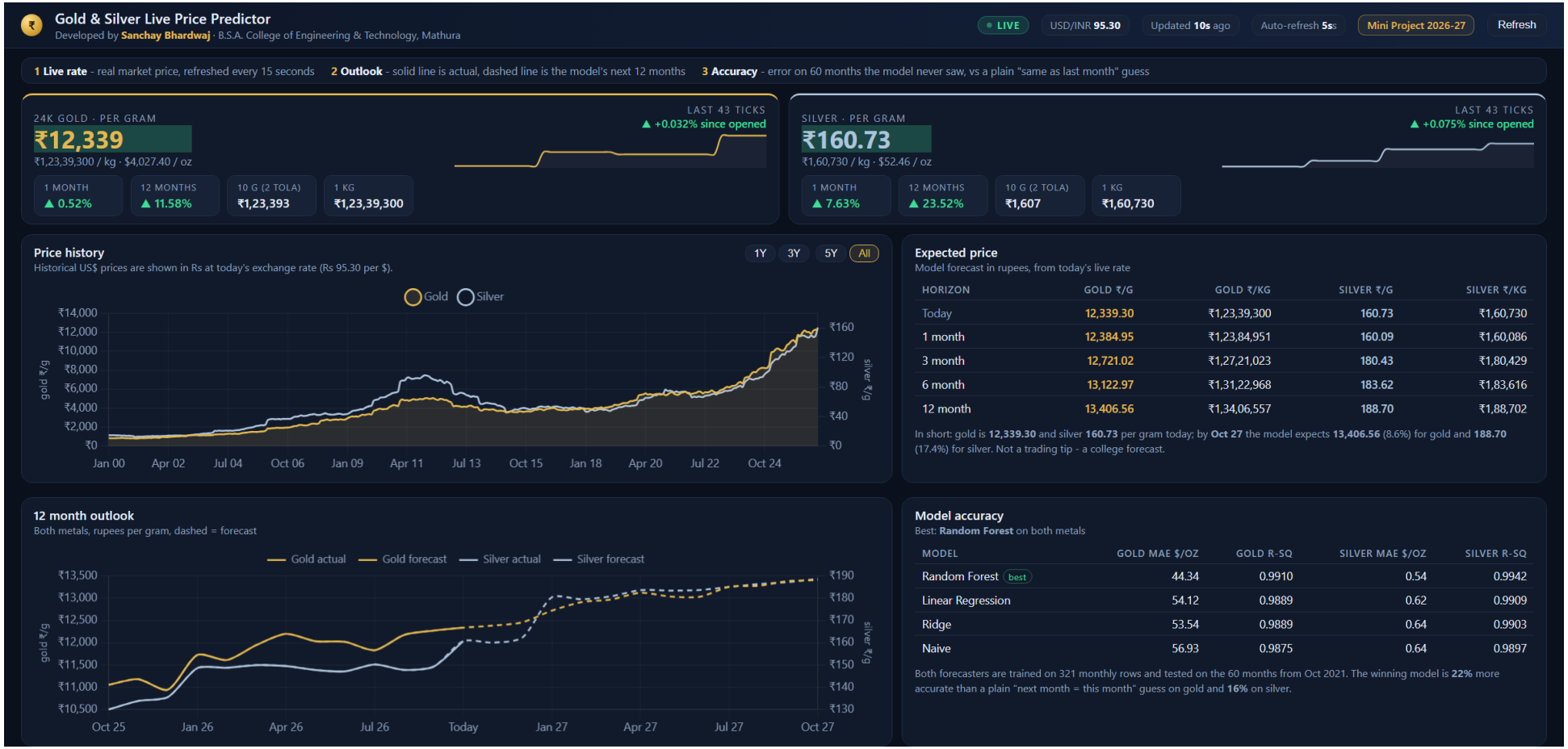
Results

Model	Gold MAE (US$/oz)	Gold R-sq	Silver MAE (US$/oz)	Silver R-sq
Random Forest	44.34	0.9910	0.54	0.9942
Linear Regression	54.12	0.9889	0.62	0.9909
Ridge	53.54	0.9889	0.64	0.9903
Naive baseline	56.93	0.9875	0.64	0.9897

Random Forest wins on both metals and beats the naive baseline. Test window: last 60 months, from Oct 2021. Lower MAE and higher R-squared are better.

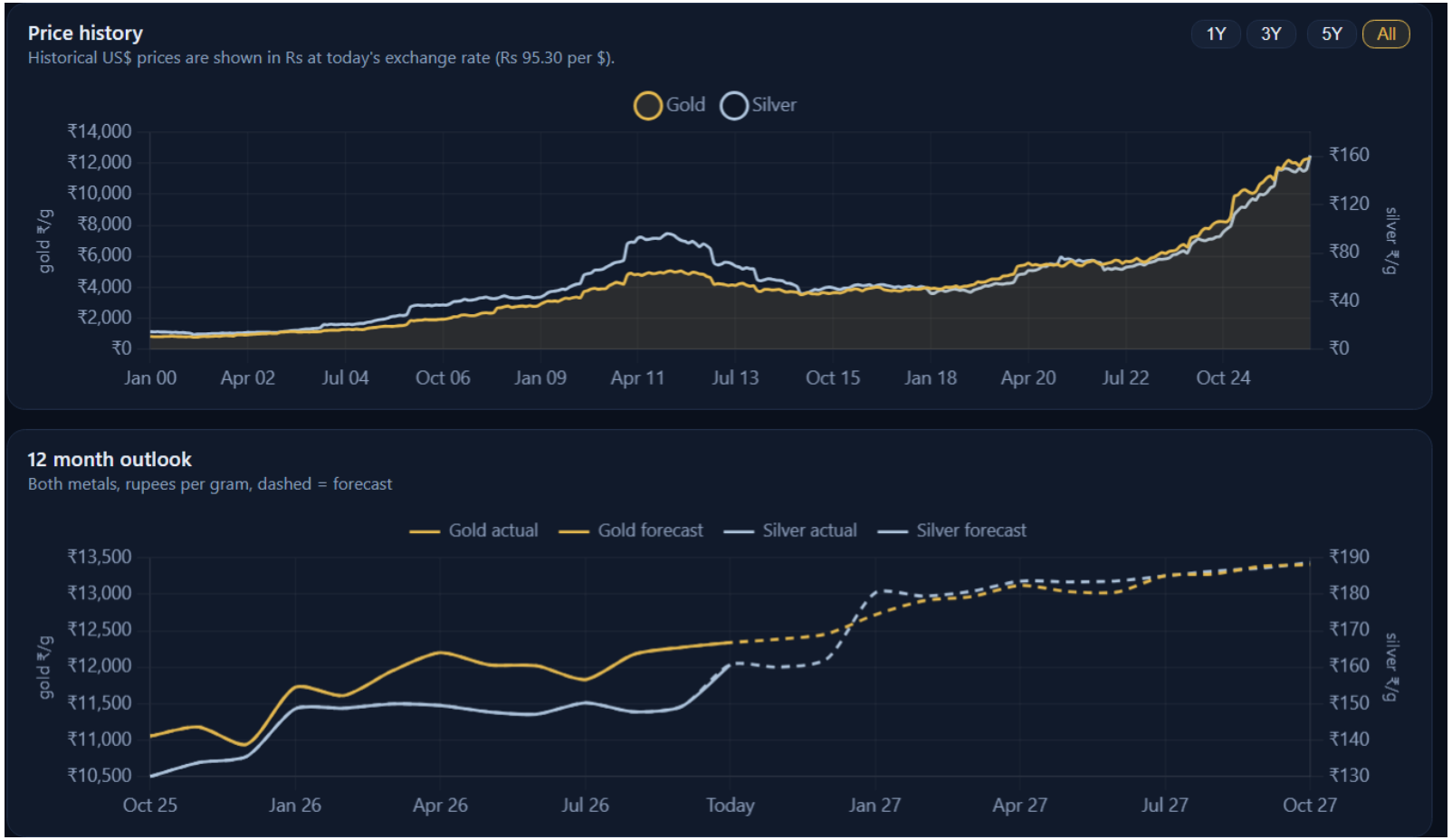


Application - Live Dashboard



The complete app on one screen: today's live gold and silver rate in rupees per gram, per 10 gram and per kg, the history since 2000, the 12 month outlook and the accuracy table.

Application - History and Forecast



Top: monthly price history of both metals since 2000. Bottom: solid line is the actual price, dashed line is what the model expects for the next 12 months.

Application - Expected Price and Accuracy

Expected price

Model forecast in rupees, from today's live rate

HORIZON	GOLD ₹/G	GOLD ₹/KG	SILVER ₹/G	SILVER ₹/KG
Today	12,339.30	₹1,23,39,300	160.73	₹1,60,730
1 month	12,384.95	₹1,23,84,951	160.09	₹1,60,086
3 month	12,721.02	₹1,27,21,023	180.43	₹1,80,429
6 month	13,122.97	₹1,31,22,968	183.62	₹1,83,616
12 month	13,406.56	₹1,34,06,557	188.70	₹1,88,702

In short: gold is **12,339.30** and silver **160.73** per gram today; by **Oct 27** the model expects **13,406.56** (8.6%) for gold and **188.70** (17.4%) for silver. Not a trading tip - a college forecast.

Model accuracy

Best: **Random Forest** on both metals

MODEL	GOLD MAE $/OZ	GOLD R-SQ	SILVER MAE $/OZ	SILVER R-SQ
Random Forest best	44.34	0.9910	0.54	0.9942
Linear Regression	54.12	0.9889	0.62	0.9909
Ridge	53.54	0.9889	0.64	0.9903
Naive	56.93	0.9875	0.64	0.9897

Both forecasters are trained on 321 monthly rows and tested on the 60 months from Oct 2021. The winning model is **22%** more accurate than a plain "next month = this month" guess on gold and **16%** on silver.

Top: expected gold and silver price after 1, 3, 6 and 12 months in rupees. Bottom: error of every model on 60 unseen months against the naive baseline.

Conclusion

- Random Forest gave the lowest error on both metals and beat the naive baseline.
- Predicting returns instead of raw prices made the models reliable on a strongly trending series.
- Live prices, rupee conversion and the forecast are combined in one self-explanatory dashboard.
- The project demonstrates a complete Machine Learning pipeline - data, model, API, interface.

Future Scope

- Add real economic features such as inflation, the dollar index and interest rates.
- Try deep-learning time-series models like LSTM for longer horizons.
- Show a confidence range around the forecast instead of a single number.
- Extend to other metals and to jewelers' rate cards used in the local market.
- Publish the dashboard online with a scheduled daily data refresh.



Thank You

Questions & suggestions are welcome

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